Bootstrap Inference for Present-Value Models with Estimated Factors[†]

Xuanbin Yang, Tingting Cheng[*]

School of Finance, Nankai University, China

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Abstract: Existing bootstrap inference for present-value models typically relies on directly observed predictors, while expected returns and dividend growth may depend on information dispersed across macroeconomic and financial variables. We extend the present-value framework by introducing common factors into its state-space representation to parsimoniously incorporate such high-dimensional information, and develop bootstrap likelihood-ratio inference with estimated factors. We establish consistency of the two-step estimator and show that replacing unobserved factors with principal-components estimates has no first-order effect on either the likelihood-ratio statistic or its innovation-bootstrap distribution. An empirical application to annual U.S. data from 1960 to 2023 illustrates the bootstrap procedure.

Keywords: Present-value models; Factor models; Bootstrap likelihood-ratio tests; Return predictability

JEL classification: C12; C32; C38; G12

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[*]Corresponding author. E-mail address: Tingting.Cheng@nankai.edu.cn.

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1 Introduction

Whether variation in aggregate valuation ratios reflects changes in expected returns or expected dividend growth is central to asset-pricing research. The Campbell–Shiller present-value relation links the price–dividend ratio to expectations of future returns and dividend growth (Campbell and Shiller 1988; Yu and Yan 2023). A prominent interpretation attributes most valuation-ratio variation to expected returns, because dividend yields forecast returns whereas aggregate dividend growth appears only weakly predictable (Fama and French 1988; Cochrane 2008; Yu and Chen 2024). Subsequent work, however, shows that this conclusion can be sensitive to the sample, forecasting specification, treatment of latent expectations, modeling of dividend dynamics, and temporal instability (Chen 2009; van Binsbergen and Koijen 2010; Chen et al. 2012; Pettenuzzo et al. 2020; Yu, Huang and Chen 2023; Yu, Chen and Li 2023). Reliable inference on both components is therefore important, especially in the short annual samples typically available for this analysis.

Latent-state present-value models provide a unified framework for jointly studying return and dividend-growth predictability, but conventional asymptotic inference can be unreliable in finite samples. Piatti and Trojani (2020) show that asymptotic likelihood-ratio (LR) tests can over-reject and propose an innovation-based bootstrap that delivers more reliable finite-sample critical values. Expected returns and expected dividend growth, however, are conditional expectations, so evidence about their predictability can depend on the information set used to model them (Lettau and Ludvigson 2005). Existing observed-predictor specifications based on a few individually selected variables may omit relevant external information and make conclusions sensitive to predictor choice. A natural way to address this information-set limitation without proliferating state-space parameters is to use a latent factor model: a small number of latent common factors can summarize the shared variation in a large panel and retain predictive signals dispersed across individual series (Kelly and Pruitt 2013; Cheng et al. 2024; Guo et al. 2026). This dimension reduction, however, creates a new inferential problem. Because the factors are unobserved and must be estimated, factor-estimation error enters both the state-space likelihood and the bootstrap procedure. The consistency of the resulting estimator and the validity of bootstrap inference therefore require separate theoretical justification.

This paper fills this gap by extending bootstrap likelihood-ratio inference to present-value models with estimated common factors. We first introduce latent common factors into the dynamics of expected returns and expected dividend growth and derive the resulting state-space representation. We then propose a two-step estimator that extracts the factors by principal components and

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estimates the state-space parameters by Kalman-filter maximum likelihood, and construct a feasible innovation-bootstrap procedure based on the estimated factors. We prove that the two-step estimator is consistent under standard approximate factor-model conditions and suitable regularity conditions for the infeasible state-space model. Under the additional rate condition $\sqrt{T}/N \to 0$, we show that replacing the rotated true factors with their estimates has no first-order effect on either the sample likelihood-ratio statistic or its innovation-bootstrap counterpart. Together, these results establish the first-order validity of likelihood-ratio and innovation-bootstrap inference with estimated factors.

We apply the procedure to annual U.S. data from 1960 to 2023, evaluating four estimated common factors and 16 observed predictors in separate specifications. Bootstrap inference retains evidence of return predictability in some estimated-factor specifications but provides no significant evidence of dividend-growth predictability, leaving intact the familiar conclusion that return predictability is stronger than dividend-growth predictability. Nevertheless, an estimated-factor specification yields a lower dividend-growth bootstrap p-value than any observed-predictor specification. This descriptive pattern suggests that common factors may summarize relevant information not captured by individual variables.

The remainder of the paper is organized as follows. Section 2 presents the benchmark present-value model, the present-value model with latent factors, and the two-step estimator. Section 3 develops bootstrap likelihood-ratio inference with estimated factors. Section 4 reports the empirical analysis, and Section 5 concludes. Derivations, estimation details, assumptions, proofs, and additional empirical results are provided in the Online Supplementary Material.

2 Present-value models and estimation

2.1 Benchmark present-value model

Let $r_{t+1} \equiv \log\{(P_{t+1} + D_{t+1})/P_t\}$ be the cum-dividend log market return and let $\Delta d_{t+1} \equiv \log(D_{t+1}/D_t)$ be aggregate log dividend growth. The log price-dividend ratio is $pd_t \equiv \log(P_t/D_t)$. Expected dividend growth and expected return, conditional on information available at time t, are $g_t \equiv E_t(\Delta d_{t+1})$ and $\mu_t \equiv E_t(r_{t+1})$, respectively, where E_t denotes conditional expectation given the date-t information set. In the benchmark present-value model, these expectation processes follow first-order autoregressions:

$$g_{t+1} = \gamma_0 + \gamma_1(g_t - \gamma_0) + \varepsilon^g_{t+1}, \tag{2.1}$$

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$$\mu_{t+1} = \delta_0 + \delta_1(\mu_t - \delta_0) + \varepsilon^{\mu}_{t+1}. \tag{2.2}$$

Dividend growth satisfies

$$\Delta d_{t+1} = g_t + \varepsilon^{d}_{t+1}. \tag{2.3}$$

The vector of shocks $\left(\varepsilon^{g}_{t+1}, \varepsilon^{\mu}_{t+1}, \varepsilon^{d}_{t+1}\right)'$ has covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_g^2 & \sigma_{g\mu} & \sigma_{gd} \\ \sigma_{g\mu} & \sigma_\mu^2 & \sigma_{\mu d} \\ \sigma_{gd} & \sigma_{\mu d} & \sigma_d^2 \end{pmatrix}. \tag{2.4}$$

Here $\sigma_{g\mu} = \sigma_g \sigma_\mu \rho_{g\mu}$, $\sigma_{\mu d} = \sigma_\mu \sigma_d \rho_{\mu d}$, and $\sigma_{gd} = \sigma_g \sigma_d \rho_{gd}$. For identification, we impose $\rho_{gd} = 0$.

The Campbell–Shiller log-linearization gives

$$pd_t = A - B_1(\mu_t - \delta_0) + B_2(g_t - \gamma_0), \tag{2.5}$$

where A, B_1, and B_2 are the present-value coefficients.

Estimation and bootstrap testing for the benchmark present-value model are studied in detail by Piatti and Trojani (2020). We therefore focus on the present-value model with latent factors in the remainder of the paper.

2.2 Present-value model with latent factors

We augment the benchmark information set with latent common factors extracted from a large panel of macroeconomic and financial variables. This parsimonious specification accommodates broad information without the parameter proliferation that would result from including all panel series directly.

Let $X_t = (X_{1t}, \ldots, X_{Nt})'$ denote an $N \times 1$ large-dimensional vector of observed macroeconomic and financial variables at date t, and let $e_t = (e_{1t}, \ldots, e_{Nt})'$ denote the corresponding idiosyncratic component. The approximate factor model is

$$X_t = \Lambda F_t + e_t, \qquad t = 1, \ldots, T, \tag{2.6}$$

where F_t is an $r \times 1$ vector of common factors and $\Lambda = (\lambda_1, \ldots, \lambda_N)'$ is the $N \times r$ loading matrix. Stacking observations over t, let $X = (X_1, \ldots, X_T)'$, $F = (F_1, \ldots, F_T)'$, and $e = (e_1, \ldots, e_T)'$. Then $X = F\Lambda' + e$. Because the panel is demeaned before factor extraction, we normalize the true factors to have zero unconditional mean.

We augment the benchmark dynamics with F_t as follows:

$$g_{t+1} = \gamma_0 + \gamma_1(g_t - \gamma_0) + \gamma_2' F_t + \varepsilon^{g}_{t+1}, \tag{2.7}$$

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$$\mu_{t+1} = \delta_0 + \delta_1(\mu_t - \delta_0) + \delta_2' F_t + \varepsilon_{t+1}^{\mu}, \tag{2.8}$$

$$F_{t+1} = \Xi F_t + \varepsilon_{t+1}^{F}, \tag{2.9}$$

where γ_2 and δ_2 are $r \times 1$ coefficient vectors and Ξ is an $r \times r$ transition matrix. Let $\Omega_F = \mathrm{Var}(\varepsilon_{t+1}^F)$, and let ρ_{gF}, $\rho_{\mu F}$, and ρ_{dF} collect the correlations of the factor shocks with ε_{t+1}^g, ε_{t+1}^{μ}, and ε_{t+1}^d, respectively. Dividend growth is

$$\Delta d_{t+1} = g_t + \varepsilon_{t+1}^{d}. \tag{2.10}$$

Define $\widetilde{g}_t \equiv g_t - \gamma_0$ and $\widetilde{\mu}_t \equiv \mu_t - \delta_0$. The Campbell–Shiller log-linearization gives

$$pd_t = A - (B_1 \widetilde{\mu}_t + B_3' F_t) + (B_2 \widetilde{g}_t + B_4' F_t)\,, \tag{2.11}$$

where the present-value coefficients are defined in Supplementary Appendix A.

2.3 Estimation

We propose the following two-step estimation procedure. The first step extracts common factors from the large panel by principal components. The second step treats the estimated factors as observed and estimates the state-space parameters by Gaussian Kalman-filter maximum likelihood.

Step 1: Factor estimation. Following Bai (2003), we estimate the factors by principal components using the demeaned and standardized panel X. Under the normalization $\widehat{F}'\widehat{F}/T = I_r$, the estimated factor matrix $\widehat{F}$ is $\sqrt{T}$ times the eigenvectors corresponding to the r largest eigenvalues of XX'. The corresponding factor loadings are $\widehat{\Lambda}' = (\widehat{F}'\widehat{F})^{-1}\widehat{F}'X = \widehat{F}'X/T$. The number of factors can be determined by the information criteria of Bai and Ng (2002).

Step 2: State-space estimation. We estimate the state-space parameters by maximizing the Gaussian Kalman-filter likelihood; see Supplementary Appendix B.

Let θ denote the parameter vector of the present-value model with latent factors, whose components are listed in Supplementary Appendix B. For any r-dimensional sequence $f = \{f_t\}_{t=1}^T$, define $Y_t(f) \equiv (\Delta d_t, pd_t, f_t')'$, and $\mathcal{Y}_T(f) \equiv \{Y_t(f)\}_{t=1}^T$. Let $\mathcal{L}_T(\theta, \mathcal{Y}_T)$ denote the Gaussian Kalman-filter log-likelihood, and write $\mathcal{L}_T(\theta; f) \equiv \mathcal{L}_T\left(\theta, \mathcal{Y}_T(f)\right)$. For the principal-components estimator $\widehat{F}_t$, there exists an invertible $r \times r$ matrix H such that $\widehat{F}_t$ estimates $H'F_t$. Since $\gamma' F_t = (H^{-1}\gamma)'H'F_t$ for any factor coefficient γ, applying the same transformation to all factor-related parameters gives the rotated true parameter vector θ^H. This reparameterization leaves likelihood values, zero restrictions, and LR statistics unchanged. Henceforth, Θ denotes the parameter space of the H-rotated representation, and all restrictions below refer to this representation. We denote the

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infeasible data vector and log-likelihood by $Y_t^H \equiv Y_t(H'F)$ and $\mathcal{L}_T(\theta; H'F)$, and their feasible counterparts by $\widehat{Y}_t \equiv Y_t(\widehat{F})$ and $\mathcal{L}_T(\theta; \widehat{F})$, respectively.

Under the maintained likelihood regularity conditions of Stoffer and Wall (1991), the infeasible estimator based on observed $H'F_t$ is consistent:

$$\widehat{\theta}^H = \arg\max_{\theta\in\Theta} \mathcal{L}_T(\theta; H'F) \to_p \theta^H.$$

Replacing $H'F_t$ with $\widehat{F}_t$ gives the feasible estimator

$$\widehat{\theta} = \arg\max_{\theta\in\Theta} \mathcal{L}_T(\theta; \widehat{F}).$$

The following theorem establishes the consistency and convergence rate of the feasible estimator.

Theorem 2.1. *Under the maintained likelihood regularity conditions of Stoffer and Wall (1991) and Assumptions A–E(i), as $N, T \to \infty$, the following results hold:*

(i) The feasible estimator is consistent:

$$\widehat{\theta} \to_p \theta^H.$$

(ii) If, in addition, Assumption E(ii)–(iv) hold and $\sqrt{T}/N \to 0$, the feasible estimator is root-T consistent:

$$\sqrt{T}(\widehat{\theta} - \theta^H) = O_p(1).$$

Corollary 2.2. *For a restricted parameter space $\Theta_0 \subset \Theta$, define $\widehat{\theta}_0 = \arg\max_{\theta\in\Theta_0} \mathcal{L}_T(\theta; \widehat{F})$. If $\theta^H \in \Theta_0$, then under the conditions of Theorem 2.1(i),*

$$\widehat{\theta}_0 \to_p \theta^H.$$

If the additional conditions in part (ii) hold, then

$$\sqrt{T}(\widehat{\theta}_0 - \theta^H) = O_p(1).$$

Together, Theorem 2.1 and Corollary 2.2 establish consistency of both the unrestricted and restricted estimators when the rotated true factors are replaced by their principal-components estimates. Under the stronger conditions, the root-T convergence rate is also preserved for both estimators. These results ensure that the feasible likelihood-based procedure inherits the consistency and local asymptotic behavior of the corresponding infeasible observed-factor procedure, which is essential for establishing the likelihood-ratio equivalence results below.

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3 Bootstrap likelihood-ratio tests

3.1 Null hypotheses

Testing return and dividend-growth predictability amounts to testing whether expected returns and expected dividend growth vary over time. In the present-value model with latent factors, the null hypothesis of no dividend-growth predictability is

$$H_0^{\text{div}}: \quad \gamma_1 = 0, \quad \gamma_2 = 0_r, \quad \sigma_g = 0, \quad \rho_{g\mu} = 0, \quad \rho_{gF} = 0_r. \tag{3.1}$$

The null H_0^{div} imposes $m_d = 2r + 3$ restrictions and implies $g_t = \gamma_0$, so expected dividend growth is constant.

Similarly, the null hypothesis of no return predictability is

$$H_0^{\text{ret}}: \quad \delta_1 = 0, \quad \delta_2 = 0_r, \quad \sigma_\mu = 0, \quad \rho_{g\mu} = 0, \quad \rho_{\mu d} = 0, \quad \rho_{\mu F} = 0_r. \tag{3.2}$$

The null H_0^{ret} imposes $m_r = 2r + 4$ restrictions and implies $\mu_t = \delta_0$, so expected returns are constant.

For each null hypothesis, let Θ_0 denote the corresponding restricted parameter space and let m denote its number of restrictions. Given a factor sequence f, the LR statistic for testing that hypothesis is

$$LR_T(f) = 2\left[\max_{\theta\in\Theta} \mathcal{L}_T(\theta; f) - \max_{\theta\in\Theta_0} \mathcal{L}_T(\theta; f)\right]. \tag{3.3}$$

Following Piatti and Trojani (2020), we maintain for the infeasible observed-factor model that, under the null hypothesis,

$$LR_T(H'F) \Rightarrow \chi^2_m, \tag{3.4}$$

and that the corresponding innovation bootstrap is first-order valid.

3.2 Bootstrap procedure

We first describe the innovation bootstrap when $H'F_t$ is observed. This infeasible procedure is the vector-predictor analogue of the bootstrap in Piatti and Trojani (2020).

Let $\mathcal{X}_t$ denote the state vector of the present-value model with latent factors. Using its state-space representation, the model can be written as

$$\mathcal{X}_{t+1} = \mathcal{F}\mathcal{X}_t + \mathcal{B}u^H_{t+1} + \Gamma\varepsilon^X_{t+1}, \tag{3.5}$$

$$Y^H_t = M_0 + M_1 Y^H_{t-1} + M_2\mathcal{X}_t, \tag{3.6}$$

where $\mathcal{F}, \mathcal{B}, \Gamma, M_0, M_1$, and M_2 are parameter-dependent matrices and $u^H_{t+1} \equiv H'F_{t-1}$. The

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innovation-form representation implied by the Kalman filter is

$$\mathcal{X}_{t+1,t} = \mathcal{F}\mathcal{X}_{t,t-1} + \mathcal{B}u_{t+1}^H + \mathcal{F}\mathcal{K}_t\eta_t^H(\theta), \tag{3.7}$$

$$Y_t^H = M_0 + M_1 Y_{t-1}^H + M_2\mathcal{X}_{t,t-1} + \eta_t^H(\theta), \tag{3.8}$$

where $\mathcal{X}_{t,t-1}$ is the one-step-ahead state prediction and $\eta_t^H(\theta) \equiv Y_t^H - M_0 - M_1 Y_{t-1}^H - M_2\mathcal{X}_{t,t-1}$ is the prediction error, $\mathcal{K}_t$ is the Kalman gain, and $S_t(\theta)$ is the innovation covariance matrix. All matrices in (3.7)–(3.8) are evaluated at the relevant parameter value.

The infeasible bootstrap LR test is based on the restricted estimator $\widehat{\theta}_0^H \equiv \arg\max_{\theta\in\Theta_0} \mathcal{L}_T(\theta; H'F)$ and is implemented as follows.

1. **Restricted standardized innovations.** Using $\widehat{\theta}_0^H$, construct

$$\widehat{e}_{0t}^H = \{S_t(\widehat{\theta}_0^H)\}^{-1/2}\eta_t^H(\widehat{\theta}_0^H), \qquad t = 1,\dots,T. \tag{3.9}$$

2. **Resampling.** Recenter the standardized innovations as $\widehat{e}_{0t}^{H,c} = \widehat{e}_{0t}^H - T^{-1}\sum_{s=1}^T \widehat{e}_{0s}^H$. Draw $I_1,\dots,I_T$ independently from the uniform distribution on $\{1,\dots,T\}$, and set $\widehat{e}_{0t}^{H,*} = \widehat{e}_{0,I_t}^{H,c}$.

3. **Bootstrap sample generation.** Using the innovation-form representation (3.7)–(3.8), generate a bootstrap sample $\{Y_t^{H,*}\}_{t=1}^T$. Specifically, for $t = 1,\dots,T$, set

$$Y_t^{H,*} = M_0 + M_1 Y_{t-1}^{H,*} + M_2\mathcal{X}_{t,t-1}^{H,*} + \{S_t(\widehat{\theta}_0^H)\}^{1/2}\widehat{e}_{0t}^{H,*}, \tag{3.10}$$

$$\mathcal{X}_{t+1,t}^{H,*} = \mathcal{F}\mathcal{X}_{t,t-1}^{H,*} + \mathcal{B}u_{t+1}^{H,*} + \mathcal{F}\mathcal{K}_t\{S_t(\widehat{\theta}_0^H)\}^{1/2}\widehat{e}_{0t}^{H,*}, \tag{3.11}$$

where all parameter-dependent matrices are evaluated at $\widehat{\theta}_0^H$. Here $u_{t+1}^{H,*} \equiv (H'F_{t-1})^*$ is obtained from the factor component of $Y_{t-1}^{H,*}$, $Y_t^{H,*} \equiv (\Delta d_t^{H,*}, pd_t^{H,*}, \{(H'F_t)^*\}')'$, and $\mathcal{Y}_T^{H,*} \equiv \{Y_t^{H,*}\}_{t=1}^T$. Initialize the recursion with $Y_0^{H,*} = Y_0^H$ and $\mathcal{X}_{1,0}^{H,*} = \mathcal{X}_{1,0}$.

4. **Bootstrap estimation.** Define $\mathcal{L}_T^{H,*}(\theta) \equiv \mathcal{L}_T(\theta, \mathcal{Y}_T^{H,*})$. The restricted and unrestricted bootstrap estimators are

$$\widehat{\theta}_0^{H,*} = \arg\max_{\theta\in\Theta_0} \mathcal{L}_T^{H,*}(\theta), \qquad \widehat{\theta}^{H,*} = \arg\max_{\theta\in\Theta} \mathcal{L}_T^{H,*}(\theta).$$

5. **Bootstrap likelihood-ratio statistic.** Compute

$$LR_T^{H,*} = 2\left[\mathcal{L}_T^{H,*}\left(\widehat{\theta}^{H,*}\right) - \mathcal{L}_T^{H,*}\left(\widehat{\theta}_0^{H,*}\right)\right]. \tag{3.12}$$

6. **Bootstrap distribution.** Repeat Steps 2–5 B times. The empirical distribution of the B statistics $LR_{T,b}^{H,*}$, $b = 1,\dots,B$, approximates the null distribution of $LR_T(H'F)$.

In practice, $H'F_t$ is unobserved. We implement the feasible bootstrap by replacing $H'F_t$ and Y_t^H in estimation with $\widehat{F}_t$ and $\widehat{Y}_t$, respectively, and using the corresponding generated variables F_t^* and

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$Y_t^* \equiv (\Delta d_t^*, pd_t^*, F_t^{*\prime})'$ in the bootstrap recursion. Here F_t^* denotes the factor component generated by the feasible recursion. Let $\mathcal{Y}_T^* \equiv \{Y_t^*\}_{t=1}^T$ and define $\mathcal{L}_T^*(\theta) \equiv \mathcal{L}_T(\theta, \mathcal{Y}_T^*)$. The corresponding restricted and unrestricted estimators are

$$\widehat{\theta}_0^* = \arg\max_{\theta \in \Theta_0} \mathcal{L}_T^*(\theta), \qquad \widehat{\theta}^* = \arg\max_{\theta \in \Theta} \mathcal{L}_T^*(\theta),$$

and the resulting statistic is denoted by LR_T^*. Thus a superscript $*$ alone denotes the feasible bootstrap, while $H,*$ denotes its infeasible counterpart based on the rotated true factors. All remaining steps, including recentering and independent resampling of the standardized innovation vectors, are unchanged. We do not resample the large panel or repeat principal-components estimation in the bootstrap.

For the following theorem and corollary, we maintain the likelihood and innovation-bootstrap regularity conditions of Stoffer and Wall (1991) for the infeasible model.

Theorem 3.1. *Suppose Assumptions A–E hold. If $N, T \to \infty$ and $\sqrt{T}/N \to 0$, then, under the null hypothesis,*

$$LR_T(\widehat{F}) - LR_T(H'F) = o_p(1). \tag{3.13}$$

Moreover,

$$LR_T^* - LR_T^{H,*} = o_{p^*}(1). \tag{3.14}$$

Theorem 3.1 shows that replacing $H'F$ with $\widehat{F}$ has no first-order effect on either the observed-sample LR statistic or its bootstrap counterpart. Combining (3.13) with (3.4) gives $LR_T(\widehat{F}) \Rightarrow \chi^2_m$. The bootstrap equivalence leads to the following validity result for the feasible procedure.

Corollary 3.2 (Feasible bootstrap validity)**.** *Under the conditions of Theorem 3.1, the feasible innovation-form bootstrap consistently estimates the null distribution of $LR_T(\widehat{F})$:*

$$\sup_x \left| P^*\{LR_T^* \le x\} - P\{LR_T(\widehat{F}) \le x\} \right| \to_p 0, \tag{3.15}$$

Corollary 3.2 carries the conditional chi-square limit of $LR_T^{H,*}$ (see Theorem 1 of Piatti and Trojani 2020) over to LR_T^* through the feasible–infeasible equivalence in Theorem 3.1. It therefore justifies inference based on the empirical distribution of LR_T^*. With B bootstrap replications, the corresponding p-value is

$$p_T^* = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\{LR_{T,b}^* > LR_T(\widehat{F})\}.$$

The null hypothesis is rejected at level α whenever $p_T^* < \alpha$.

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4 Empirical analysis

4.1 Empirical design

We obtain monthly returns with and without dividends on the value-weighted portfolio of all New York Stock Exchange, Amex, and Nasdaq stocks from January 1959 through December 2023 from the Center for Research in Security Prices (CRSP). We construct annual time series of aggregate dividends and prices, assuming that monthly dividends are cash-reinvested at the 30-day T-bill rate, and retain the period 1960–2023. Data on 30-day T-bill rates are also obtained from CRSP.

We extract 4 latent factors by principal components from 226 quarterly FRED-QD series, annualized by calendar-year averaging and then transformed to stationarity (McCracken and Ng 2021). The number of factors is selected using the information criteria of Bai and Ng (2002). The predictor set also includes 16 variables from the updated data set of Welch and Goyal (2008). Supplementary Table F.1 provides their definitions. All candidate predictors are standardized before estimation.

Although the theoretical model permits several factors to enter jointly, each empirical specification adds one variable at a time to the benchmark model: either one of the 4 estimated factors or one of the 16 observed predictors, thereby avoiding parameter proliferation in the short annual sample. We first estimate the unconstrained benchmark model and compute $R^2_{\text{ret}} = 1 - \text{Var}(r_{t+1} - \widehat{\mu}_t)/\text{Var}(r_{t+1})$ and $R^2_{\text{div}} = 1 - \text{Var}(\Delta d_{t+1} - \widehat{g}_t)/\text{Var}(\Delta d_{t+1})$, where $\widehat{\mu}_t$ and $\widehat{g}_t$ are the estimated conditional means from the unconstrained model. We then test separately the null hypotheses of no dividend-growth predictability and no return predictability, reporting asymptotic and bootstrap LR p-values. Bootstrap results use $B = 200$ replications.

4.2 Empirical results

Table 4.1 reports the unconstrained benchmark estimates together with estimates under no return predictability and no dividend-growth predictability. The table also reports the two model-implied R^2 measures, the maximized log likelihood, the LR statistic, and its asymptotic and bootstrap p-values. The unconstrained estimates imply $R^2_{\text{ret}} = 6.84\%$ and $R^2_{\text{div}} = 15.29\%$. The asymptotic tests reject both null hypotheses, with p-values below 0.01% for no return predictability and 1.64% for no dividend-growth predictability. The corresponding bootstrap p-values increase to 5.50% and 15.00%, so neither null is rejected at the 5% level in the benchmark model. This contrast

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is consistent with the finite-sample over-rejection documented by Piatti and Trojani (2020).

Table 4.1: Benchmark-model estimates and LR tests

	Unconstrained	No return predictability	No dividend-growth predictability
γ_0	0.058	0.062	0.057
δ_0	0.081	0.085	0.080
γ_1	0.269	0.988	0
δ_1	0.917	0	0.907
σ_g	0.065	0.006	0
σ_μ	0.017	0	0.020
σ_d	0.001	0.069	0.067
$\rho_{g\mu}$	0.248	0	0
$\rho_{\mu d}$	−0.909	0	0.348
R^2_{ret}	6.84%	0.00%	7.26%
R^2_{div}	15.29%	−3.96%	0.00%
LogL	226.44	213.41	221.31
LR	–	26.07	10.27
Asym. p-value	–	$< 0.01\%$	1.64%
Boot. p-value	–	5.50%	15.00%

Notes: LogL denotes the maximized Gaussian Kalman-filter log-likelihood. LR is the likelihood-ratio statistic; asymptotic and bootstrap p-values are reported in percentage terms.

Supplementary Table F.2 reports the unconstrained estimates and R^2 values for all 20 candidates, 15 of which increase both model-implied R^2 values relative to the benchmark. Among these 15 candidates, Table 4.2 reports the two observed predictors and the estimated factor with the largest model-implied R^2_{div}; Table 4.3 applies the same selection rule using R^2_{ret}. Supplementary Tables F.3 and F.4 report all 15 candidates.

At the 5% significance level, 10 of the 15 asymptotic tests reject the null of no dividend-growth predictability, whereas none of the bootstrap tests do. Among the 15 retained specifications, F1 produces the smallest bootstrap p-value, 15.50%, which is still well above conventional significance levels. Thus, expanding the information set provides no statistically significant evidence of dividend-growth predictability.

For return predictability, all 15 asymptotic tests reject the null, whereas only 9 bootstrap tests do so, including those for F1 and F4. Bootstrap inference therefore weakens the asymptotic evidence but does not eliminate it. Although SVAR produces the second-largest increase in R^2_{ret} among the retained observed predictors, its bootstrap p-value is 11.50%, exceeding the 5.50% benchmark value. Observable predictors can therefore weaken, rather than reinforce, the evidence once finite-sample uncertainty is taken into account.

Overall, the high-dimensional information set does not overturn the familiar conclusion within the present-value framework that return predictability is stronger than dividend-growth predictability. At the same time, the fact that F1 delivers a lower dividend-growth bootstrap p-value than any

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Table 4.2: Test of no dividend-growth predictability

$H_0^{\text{div}} : \gamma_1 = \gamma_2 = \sigma_g = \rho_{g\mu} = \rho_{gz} = 0$

	TMS	DFY	F1
Panel A: Constrained maximum-likelihood estimates			
γ_0	0.057	0.057	0.058
δ_0	0.080	0.080	0.081
γ_1	0	0	0
δ_1	0.898	0.910	0.907
σ_g	0	0	0
σ_μ	0.026	0.037	0.035
σ_d	0.067	0.067	0.067
$\rho_{g\mu}$	0	0	0
$\rho_{\mu d}$	0.056	0.169	−0.335
ξ_1	0.622	0.433	0.137
ρ_{gz}	0	0	0
$\rho_{\mu z}$	0.561	0.900	0.842
ρ_{dz}	−0.373	0.007	−0.516
σ_z	0.812	0.899	0.977
δ_2	−0.008	−0.015	−0.032
γ_2	0	0	0
Panel B: Asymptotic test			
LR	10.20	15.52	12.49
p-value (%)	6.97	0.84	2.86
Panel C: Bootstrap test			
p-value (%)	24.00	19.50	15.50

Notes: z denotes the additional variable (an observed predictor or an estimated factor). TMS denotes the term spread, DFY the default yield spread, and F1 the first estimated factor. Bootstrap p-values are based on 200 replications.

Table 4.3: Test of no return predictability

$H_0^{\text{ret}} : \delta_1 = \delta_2 = \sigma_\mu = \rho_{g\mu} = \rho_{\mu d} = \rho_{\mu z} = 0$

	SVAR	EQIS	F4
Panel A: Constrained maximum-likelihood estimates			
γ_0	0.069	0.059	0.058
δ_0	0.088	0.084	0.083
γ_1	0.986	0.996	0.995
δ_1	0	0	0
σ_g	0.013	0.008	0.006
σ_μ	0	0	0
σ_d	0.067	0.068	0.068
$\rho_{g\mu}$	0	0	0
$\rho_{\mu d}$	0	0	0
ξ_1	0.309	0.746	0.731
ρ_{gz}	−0.869	−0.805	0.576
$\rho_{\mu z}$	0	0	0
ρ_{dz}	−0.259	−0.180	0.310
σ_z	0.955	0.712	0.738
δ_2	0	0	0
γ_2	0.006	0.003	−0.002
Panel B: Asymptotic test			
LR	20.75	32.93	24.39
p-value (%)	0.20	< 0.01	0.04
Panel C: Bootstrap test			
p-value (%)	11.50	1.50	3.00

Notes: z denotes the additional variable (an observed predictor or an estimated factor). SVAR denotes stock variance, EQIS equity issuance, and F4 the fourth estimated factor. Bootstrap p-values are based on 200 replications.

observed-predictor specification suggests that estimated factors can partly mitigate the informational limitation of individual predictors by summarizing a broad panel without proliferating state-space parameters.

5 Conclusions

This paper extends present-value predictability analysis beyond narrow information sets based on individual observed predictors by introducing common factors to parsimoniously summarize a large panel of macroeconomic and financial variables. We establish the consistency of the two-step estimator and show that factor estimation has no first-order effect on either the likelihood-ratio statistic or its innovation-bootstrap distribution. Applied to annual U.S. data from 1960 to 2023, the proposed bootstrap inference weakens the evidence of dividend-growth predictability while retaining evidence of return predictability in some specifications. Overall, the results illustrate how estimated factors can expand the information set in present-value models without invalidating bootstrap inference.

References

Bai, J. (2003), 'Inferential theory for factor models of large dimensions', *Econometrica* **71**(1), 135–171.

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Preprint not peer reviewed

Bai, J. and Ng, S. (2002), 'Determining the number of factors in approximate factor models', *Econometrica* **70**(1), 191–221.

Campbell, J. Y. and Shiller, R. J. (1988), 'The dividend-price ratio and expectations of future dividends and discount factors', *Review of Financial Studies* **1**(3), 195–228.

Chen, L. (2009), 'On the reversal of return and dividend growth predictability: A tale of two periods', *Journal of Financial Economics* **92**(1), 128–151.

Chen, L., Da, Z. and Priestley, R. (2012), 'Dividend smoothing and predictability', *Management Science* **58**(10), 1834–1853.

Cheng, T., Yang, X. and Zhao, A. B. (2024), 'Predicting the predictable: Decomposing and forecasting stock returns in a data-rich environment', *SSRN Electronic Journal* .

Cochrane, J. H. (2008), 'The dog that did not bark: A defense of return predictability', *Review of Financial Studies* **21**(4), 1533–1575.

Fama, E. F. and French, K. R. (1988), 'Dividend yields and expected stock returns', *Journal of Financial Economics* **22**(1), 3–25.

Guo, M., Xu, Q. and Yang, X. (2026), 'Market expectations in the cross-section of leading present values', *SSRN Electronic Journal* .

Kelly, B. and Pruitt, S. (2013), 'Market expectations in the cross-section of present values', *Journal of Finance* **68**(5), 1721–1756.

Lettau, M. and Ludvigson, S. C. (2005), 'Expected returns and expected dividend growth', *Journal of Financial Economics* **76**(3), 583–626.

McCracken, M. W. and Ng, S. (2021), 'FRED-QD: A quarterly database for macroeconomic research', *Federal Reserve Bank of St. Louis Review* **103**(1), 1–44.

Pettenuzzo, D., Sabbatucci, R. and Timmermann, A. (2020), 'Cash flow news and stock price dynamics', *Journal of Finance* **75**(4), 2221–2270.

Piatti, I. and Trojani, F. (2020), 'Dividend growth predictability and the price–dividend ratio', *Management Science* **66**(1), 130–158.

Stoffer, D. S. and Wall, K. D. (1991), 'Bootstrapping state-space models: Gaussian maximum likelihood estimation and the Kalman filter', *Journal of the American Statistical Association* **86**(416), 1024–1033.

van Binsbergen, J. H. and Koijen, R. S. J. (2010), 'Predictive regressions: A present-value approach', *Journal of Finance* **65**(4), 1439–1471.

Welch, I. and Goyal, A. (2008), 'A comprehensive look at the empirical performance of equity premium prediction', *Review of Financial Studies* **21**(4), 1455–1508.

Yu, D. and Chen, L. (2024), 'Local predictability of stock returns and cash flows', *Journal of Empirical Finance* **77**, 101485.

Yu, D., Chen, L. and Li, L. (2023), 'Time-varying predictability of the long horizon equity premium based on semiparametric regressions', *Economics Letters* **224**, 111033.

Yu, D., Huang, D. and Chen, L. (2023), 'Stock return predictability and cyclical movements in valuation ratios', *Journal of Empirical Finance* **72**, 36–53.

Yu, D. and Yan, Y. (2023), 'Joint dynamics of stock returns and cash flows: A time-varying present-value framework', *Financial Management* **52**(3), 513–541.

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This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7273870